

CFD-VisQA Benchmark Structure

Axis 1: Scenarios

S1 Heated plate

S2 Channel (lam)

S3 Channel (turb)

S4 Backward step

S5 Lid-driven cavity

S6 Heated cavity

S7 Mixed convection

S8 Heated obstacle

Dataset Statistics

S9 Rayleigh-Benard

60 CFD cases

S10 Ventilated room

258 images

279 QA questions

6 error types × 3 severities

Axis 2: Visualization

V1 Velocity contour

V2 Temperature contour

V3 Pressure contour

V4 Vector field

V5 Streamlines

V6 Composite

Axis 3: Question Level

L1 Visual reading

L2 Pattern recognition

L3 Physical reasoning ★

L4 Quantitative estimation

L5 Comparative judgment

Error Types (per scenario: 2 correct + 4 errors)

E1 Under-converged

E2 BC swap

E3 Wrong viscosity

E4 Wrong turb model

E5 Coarse mesh

E8 Gravity flip